ha-coap-node Power Review

Where the battery current goes on the `hacoap` board, and what the firmware can do about it

Repository	`ha-coap-server`, branch `claude` at commit `ed7b8c8`
Target	ha-coap-node Rev 2.0 (ACN52840 / nRF52840), Thread MTD sleepy end device, battery build (`overlay-mtd.conf` + `boards/usb.overlay`)
SDK	nRF Connect SDK v2.4.3
Review date	2026-10-24
Status	**Proposed; no changes applied yet.**
Verification	Driver behavior was checked in the NCS 2.4 sources. Current figures come from Nordic's measurements and datasheets, or are estimates where marked. **Nothing here was measured on the board.** Confirm with a Power Profiler Kit II before relying on any number.
Prepared by	Claude (AI-assisted review, prepared with Claude Code). Companion documents: `docs/hardware/ha-coap-node-rev2.0-schematic-review.pdf` and `docs/firmware/ha-coap-server-code-review.pdf`.

Summary

The firmware is already in the tens of microamps. The board's always-on hardware loads draw about **1.4 mA**, roughly 60 times more, so the largest saving is a hardware change. On the firmware side, three items are worth doing: status LEDs that can stay on solid, a button interrupt mode that may add about 20 µA in sleep, and the 1 s Thread poll period. The USB port costs nothing on battery.

#	Priority	Area	Finding	Approx. saving
H	HIGH	Hardware	Power LED, SW2/R36 and the TPS2115A draw current all the time.	~1.35 mA
1	HIGH	LEDs	The RGB LED can stay on solid for hours: off network, waiting for SRP, after a sensor error.	mA, while it happens
2	MEDIUM	GPIO	The user-button edge interrupt uses a high-accuracy GPIOTE channel.	~20 µA (reported, unverified)
3	MEDIUM	Thread	A 1 s poll period costs about 18 µA.	9–14 µA at 2–5 s
4	LOW	RAM	The RAM power-down library is enabled but never used.	~1 µA
5	LOW	Sensors	The soil sensor is powered for 200 ms on every `data` request.	~1 µA at HA's rate
U	INFO	USB	The USB port draws nothing from the battery. No change needed.	none

Current budget

Sleepy-device figures are from Nordic's measurements in the SDK (`external/nrf/doc/nrf/protocols/thread/overview/power_consumption.rst`). They were taken on an nRF52840 DK at 3.0 V and 0 dBm with a 1 s poll period, and give 3.05 µA average sleep, 18.02 µC per data poll and 1263 µC per minute (about 21 µA). Hardware figures are from the schematic review.

Load	Current	Kind	Basis
Green power LED (LED1)	~1 mA	Hardware	Schematic review
SW2 in the MTD position, through R36	~300 µA	Hardware	Schematic review
TPS2115A power mux quiescent	~55 µA typ.	Hardware	TPS2115A datasheet
Thread data polls, 1 s period	~18 µA	Firmware	Nordic measurement
User button GPIOTE IN channel	~20 µA?	Firmware	Reported for nRF52; unverified on this board
nRF52840 sleep floor, RAM partly powered down	~3 µA	Firmware	Nordic measurement
Extra RAM retention (library unused)	~1 µA	Firmware	Estimate
Soil, air and fuel-gauge reads every 15 min	~1 µA	Firmware	Estimate; sensor current assumed ~5 mA

> **Illustration only, per 1000 mAh of battery and ignoring self-discharge:** about 1.4 mA lasts about 30 days. About 45 µA, with the hardware loads removed and no firmware changes, lasts about 2.5 years. About 10 µA, with items 2–4 applied as well, lasts several years, at which point battery self-discharge dominates.

Hardware (largest saving)

H **Always-on loads on the carrier board** HIGH

- **Power LED, ~1 mA.** Leave it unpopulated on battery boards, or fit a much larger resistor. A high-efficiency LED at 50–100 µA is still visible.
- **SW2 / R36, ~300 µA in the MTD position.** Rework the strap so that neither position conducts continuously, for example with a high-value pull resistor read once at boot. The firmware doesn't read P0.24 today.
- **TPS2115A, ~55 µA.** This is inherent to the part. A low-quiescent ideal-diode or load-switch arrangement would remove most of it in the next revision.

These come from the Rev 2.0 schematic review and are not firmware fixes. Firmware tuning below about 50 µA will not show up in battery life until they are addressed.

Firmware

1 RGB LED left on solid HIGH

Where. `src/coap_server.c`, `on_thread_state_changed()` and `blink_error()`.

- Detached or disabled: red and green stay on. This happens on a board that hasn't been commissioned yet, and on every board while the border router or Home Assistant is down.
- Attached but SRP not yet registered: blue stays on. If SRP registration never succeeds, it stays on indefinitely.
- After a sensor error, `blink_error()` leaves red on until the device attaches.

Impact. Each LED is likely a few mA, which is more than the rest of the firmware combined. The worst case is exactly when the network is down and the device can't report its battery level.

Change. Replace solid states with a short flash, for example 20 ms every 5 s, driven by a `k_timer`. Optionally stop indicating after a few minutes.

Test. Power the border router off and check the LED pattern and the average current.

2 User button interrupt uses a high-accuracy GPIOTE channel MEDIUM

Where. `usr_button_init()` calls `gpio_pin_interrupt_configure_dt(..., GPIO_INT_EDGE_TO_ACTIVE)` on P1.13.

Evidence. In `external/zephyr/drivers/gpio/gpio_nrfx.c` (lines 254–268), an edge interrupt allocates a GPIOTE IN channel unless the pin is listed in the port's `sense-edge-mask`. On nRF52 parts this mode is reported to add about 20 µA in System ON sleep, against about 2 µA with the PORT/SENSE event. That figure comes from community and Nordic forum reports and has not been measured on this board.

Change. Add `sense-edge-mask = <(1 << 13)>;` to `&gpio1` in `hacoap.dts`. The code stays the same. A button press is far longer than the PORT event's latency, so nothing is lost.

Test. Measure sleep current before and after, and check that the button still starts the pump.

3 Thread poll period MEDIUM

Where. `overlay-mtd.conf`: `CONFIG_OPENTHREAD_POLL_PERIOD=1000`.

Evidence. Average current is about `3 µA + 18 µC ÷ period`:

Poll period	1 s	2 s	3 s	5 s	10 s
Average current	~21 µA	~12 µA	~9 µA	~6.6 µA	~4.8 µA

Trade-off. A message for the device waits at its parent until the next poll, so latency equals the poll period. The integration sends pump and ping commands as confirmable (CON) PUTs, and aiocoap retransmits after roughly 2–3 s. A 2–3 s period is therefore safe. Longer periods work but cause duplicate retransmissions and slower pump response, so test them with Home Assistant first.

Not recommended. Thread 1.2 CSL (synchronized sleepy end device) saves only about 12% at the same period in Nordic's measurements (1112.5 vs 1263 µC per minute), and it needs CSL support in the border router.

4 RAM power-down library enabled but unused LOW

Where. `overlay-mtd.conf` sets `CONFIG_RAM_POWER_DOWN_LIBRARY=y`, but nothing calls `power_down_unused_ram()`, so every RAM section stays powered. Nordic's samples call it explicitly.

Change. Add `CONFIG_RAM_POWER_ADJUST_ON_HEAP_RESIZE=y` (experimental in NCS 2.4). It powers down unused RAM at boot and powers sections back up as the newlib heap grows. Do not call `power_down_unused_ram()` directly: the newlib heap lives in that RAM, and newlib's floating-point formatting can allocate from it.

Impact. About 1 µA. The image uses about 97 KB of the 256 KB RAM.

5 Sensor power per `data` request LOW

Where. `read_soil_moisture()` powers the soil sensor for `SENSOR_POWER_UP_TIME` (200 ms) on every request.

Impact. At an assumed ~5 mA for a capacitive sensor, that is about 1 mC per read. The HA integration requests `data` every 15 min (`CONST_DATA_SCAN_PERIOD_S = 900`), `info` every hour and `ping` every hour, so the average is about 1 µA. No change is needed unless the poll rate goes up.

Optional. Measure the sensor's settling time and shorten `SENSOR_POWER_UP_TIME` to match. If HA ever polls faster, read the sensors on a slow local timer and answer from cached values.

USB port

U USB costs nothing on battery INFO

Unplugged. The Zephyr driver (`external/zephyr/drivers/usb/device/usb_dc_nrfx.c`, lines 674–702) enables the USBD peripheral and requests the 32 MHz crystal only when VBUS is detected, and disables both when VBUS is removed. With no cable, the USB block is off and draws essentially nothing. Keeping USB in the battery build costs flash, not current, and it keeps mcumgr recovery over USB available.

Plugged in. From the nRF52840 Product Specification (USBD electrical specification):

Parameter	Condition	Typical
$I_{USB,ACTIVE,VBUS}$	From VBUS, USB active	2.4 mA
$I_{USB,ACTIVE,VDD}$	From VDD, CPU running, USB active, LDO	7.73 mA
$I_{USB,SUSPEND,VBUS}$	From VBUS, USB suspended, CPU sleeping	262 µA
$I_{USB,SUSPEND,VDD}$	From VDD, USB suspended, CPU sleeping	173 µA
$I_{USB,DISABLED,VDD}$	From VDD, USB disabled, VBUS connected, CPU sleeping	7 µA

On this board the TPS2115A runs the system from USB whenever a cable is connected, so none of this current comes from the battery.

Already in good shape

- DC/DC regulator enabled (`CONFIG_BOARD_ENABLE_DCDC`, default `y`).
- UART0 disabled in the battery build, so its receiver isn't running in sleep.

- The LSM6DSM is left powered down: the driver's accelerometer and gyroscope output data rates are 0.
- The VL53L0X shutdown pin is left to its pull-up, with the sensor in standby rather than pulling ~300 µA through R30.
- The soil sensor is power-gated through the FPF1320 between reads.
- The watchdog feed wakes the CPU every 10 s, which costs well under 0.1 µA.
- The MAX17048 hibernates automatically at low battery current with its default thresholds.

How to measure

1. Power the board from a Power Profiler Kit II in source mode at the battery voltage, connected to the battery connector, with USB unplugged.
2. Capture at least 60 s after the device has attached and SRP has registered. The steady pattern is sleep plus one data poll per period.
3. Trigger a `data` request and a pump command during the capture to see their charge.
4. Repeat with the power LED removed and SW2 in each position to separate hardware from firmware current.

References

- Nordic, OpenThread power consumption (NCS 2.4): `external/nrf/doc/nrf/protocols/thread/overview/power_consumption.rst`
- nRF52840 Product Specification, USBD electrical specification: https://docs.nordicsemi.com/bundle/ps_nrf52840/page/usbd.html
- GPIOTE IN versus PORT sleep current: https://github.com/adafruit/circuitpython/pull/9534 and https://devzone.nordicsemi.com/f/nordic-q-a/86598/reduce-gpio-current-draw/362651
- Zephyr sources: `drivers/usb/device/usb_dc_nrfx.c`, `drivers/gpio/gpio_nrfx.c`; NCS `lib/ram_pwrdn`
- HA integration scan periods: yanntalhouarne/ha-coap-integration, `sensor.py` and `binary_sensor.py`